

Towards Realistic Virtual Try-on for E-commerce by Sewing Pattern Estimation

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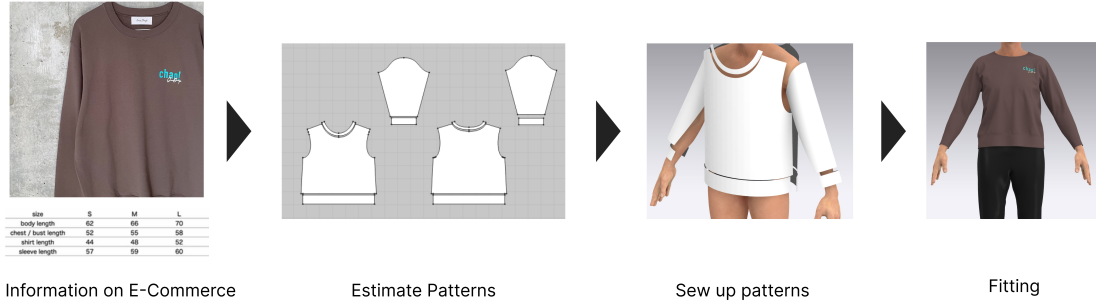


Figure 1: Pattern making pipeline overview.

CCS CONCEPTS

- **Software and its engineering** → **E-commerce infrastructure**;
- **Computing methodologies** → *Parametric curve and surface models*.

KEYWORDS

Parametric design, Garment pattern, Virtual try-on

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1 INTRODUCTION

Virtual try-on systems enable consumers to try products and makeup using their devices, offering a potential solution to online-specific problems in the fashion industry, such as returns due to incorrect sizing. However, there are challenges in providing meaningful try-on experiences, including accurately representing size differences and efficiently generating 3D models for a wide variety of products.

NeuralTailor [Korosteleva and Lee 2022] is the most similar approach to ours, using sewing patterns as a base representation of garments. However, it reconstructs sewing patterns from a point

cloud of the garment 3D model, which is normally not available in e-commerce contexts. Other studies use parametric patterns to generate 3D models, such as Detailed Garment Recovery [Yang et al. 2016], which creates 3D models of human bodies and clothing from single photographs but does not reflect actual garment sizes and takes over 4 hours to generate a model. Kang [2021] used body size information as input to create patterns that fit the body but do not satisfy our problem settings due to generating speed and actual garment size concerns.

In this paper, we propose a pipeline processing framework that traces the physical pattern-based dressmaking flow to express a realistic clothes size to address these challenges. The first step is predicting a sewing pattern automatically by utilizing information available on an online shop's item page, e.g. images of items and measurements. Then our system sews up the sewing pattern and makes a 3D model that is ready for fitting simulation. Finally, it renders photo-realistic fitting images by running cloth simulation.

2 METHOD

2.1 Problem Setting

We developed a system tailored for use in general fashion e-commerce platforms, assuming that the features are provided on widely recognized websites. They typically offer basic measurements (e.g., inseam, rise, hip, and waist for pants) and images for almost every item.

Our objective is to create an automatic virtual fitting system that relies solely on these sources of information. Given the extensive variety of clothing options, the system should allow users to effortlessly try on new garments as they become available. This requirement ensures that the system operates effectively using only the information provided by online fashion retailers.

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Figure 2: Simulated clothes(left) , simulated clothes with transparent texture(middle), and acutal clothes(right).

2.2 System

Figure 1 presents an overview of our system, which consists of a pipeline structure connecting three processes.

Sewing pattern prediction. A sewing pattern is a set of instructions for creating a specific garment, consisting of 2D fabric pieces and information on stitching them together to form the actual garment. Sewing patterns are almost always required when producing real clothing.

In this step, we predict the sewing pattern using the provided measurements and images. Manually drawing a pattern from this information typically requires apparel expertise. In our approach, we define a set of functions that output the coordinates of sewing pattern outline points and parameters of the lines connecting them, taking the given basic measurements as arguments. The target sewing pattern is represented as a set of points, straight lines, and quadratic Bézier curves.

It is important to note that reconstructing a sewing pattern from basic measurements is not a deterministic process. The system generates the most reasonable pattern based on the available information, producing a rough sewing pattern that satisfies the given conditions while disregarding minor shape differences.

3D model preparation by sewing up. This step involves creating a garment 3D model ready for subsequent cloth simulation. Stitching is the process of connecting corresponding points and lines between each garment piece. Automating this process is possible by specifying the stitching locations in advance according to the clothing type. For this project, we utilized Blender’s¹ stitching function and automated it with scripts. Consequently, we converted the 2D patterns into a 3D model suitable for simulation.

Fitting simulation. To demonstrate the virtual try-on, the system places the created clothing model on a 3D human avatar and uses CLO² to perform a realistic cloth simulation. The 3D human avatar is customized to reflect the user’s body shape. In this study, we adjust the avatar’s height, weight, and gender.

In addition to displaying the clothing’s appearance, the system uses a transparent texture to visualize the tightness or looseness of the garment on the user’s body. This provides valuable feedback to users when deciding which size to purchase.



Figure 3: Simulated clothes of four categories.

3 RESULTS

Figure 2 displays the fitting images rendered by our system. Figure 3 shows the rendered results of four garment categories: t-shirts, shirts, pants, and skirts.

By utilizing a sewing pattern as the fundamental representation of the garment model, we can convey the sensation of a try-on through a realistic cloth simulation. Additionally, our system accurately represents the length of clothing for each available size. These results suggest that our 3D garment model reconstruction is sufficiently accurate for virtual try-on purposes.

4 CONCLUSION

In this study, we proposed a pattern-based virtual try-on system designed for integration with e-commerce platforms. By constructing garments from patterns, we achieved a realistic appearance with adequate quality for try-on purposes. In future work, we plan to estimate dimensions not listed on e-commerce websites based on images. Furthermore, we aim to detect and apply basic colors and logos to designated areas using state-of-the-art instance segmentation technology. Additionally, we intend to consolidate all operations within Unreal Engine³ to improve the quality and speed of automation.

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¹<https://blender.jp/>

²<https://www.clovirtualfashion.com/>

³<https://www.unrealengine.com/>